

Methods and Properties

1) Create an Employee class with the following specifications:

- 1. Employee Name as string.**
- 2. Basic Salary, HRA, DA, TAX, Gross Pay and Net Salary as decimal.**
- 3. Calculate the HRA (15% of Basic Salary), DA (10% of Basic Salary), Gross Pay (Basic Salary + HRA + DA), Tax (8% of Gross Pay) and Net Pay (Gross Pay – Tax).**
- 4. A Constructor to allow to define the Employee Name and Basic Salary.**
- 5. A method CalculateNetPay() to calculate the HRA, DA, Tax, Gross Pay and Net Pay values.**
- 6. A method Display() to display the salary structure.**

using System;

class Employee

{

string empName;

decimal basicSalary;

decimal hra, da, tax, grossPay, netPay;

public Employee(string name, decimal salary)

{

```
empName = name;
basicSalary = salary;
}

public void CalculateSalary()
{
    hra = basicSalary * 15 / 100;
    da = basicSalary * 10 / 100;
    grossPay = basicSalary + hra + da;
    tax = grossPay * 8 / 100;
    netPay = grossPay - tax;
}

public void Display()
{
    Console.WriteLine("\n----- Salary Details -----");
    Console.WriteLine("Employee Name : " + empName);
    Console.WriteLine("Basic Salary : " + basicSalary);
    Console.WriteLine("HRA      : " + hra);
    Console.WriteLine("DA      : " + da);
    Console.WriteLine("Gross Pay   : " + grossPay);
    Console.WriteLine("Tax       : " + tax);
    Console.WriteLine("Net Pay    : " + netPay);
}
```

```

static void Main()
{
    Console.Write("Enter Employee Name: ");
    string name = Console.ReadLine();

    Console.Write("Enter Basic Salary: ");
    decimal salary = Convert.ToDecimal(Console.ReadLine());

    Employee emp = new Employee(name, salary);
    emp.CalculateSalary();
    emp.Display();
}
}

```

OUTPUT:

Enter Employee Name: Nandini

Enter Basic Salary: 50000

----- Salary Details -----

Employee Name : Nandini

Basic Salary : 50000

HRA : 7500

DA : 5000

Gross Pay : 62500

Tax : 5000

Net Pay : 57500

2) Create a new class called "Stock" which contains the following:

- 1. A String for the stock name.**
- 2. A String for the stock symbol.**
- 3. A double storing the previous closing price of the stock.**
- 4. A double storing the current closing price of the stock.**
- 5. A Constructor that allows you to define a stock's name, symbol, previous closing price and current closing price.**
- 6. A method GetChangePercentage() that returns the percentage change from the previous closing price to the current closing price.**

using System;

class Stock

{

string name, symbol;

double previousPrice, currentPrice;

public Stock(string n, string s, double p, double c)

{

name = n;

symbol = s;

previousPrice = p;

currentPrice = c;

}

public void Display()

```
{  
    double percentage = ((currentPrice - previousPrice) * 100) / previousPrice;  
  
    Console.WriteLine("Stock Name    : " + name);  
    Console.WriteLine("Stock Symbol  : " + symbol);  
    Console.WriteLine("Previous Price : " + previousPrice);  
    Console.WriteLine("Current Price  : " + currentPrice);  
    Console.WriteLine("Change Percentage: " + percentage.ToString("F2") +  
"%");  
}  
  
static void Main()  
{  
    Stock stock = new Stock("Reliance", "RELIANCE", 2500, 2625);  
  
    stock.Display();  
}  
}
```

OUTPUT:

Stock Name : Reliance
Stock Symbol : RELIANCE
Previous Price : 2500
Current Price : 2625
Change Percentage: 5.00%

3) Create a new class called "RandomHelper" which contains the following:

- 1. A static method called RandInt() that accepts two integers and returns a random integer between them. The numbers should be inclusive (i.e., if you call RandInt(1,10) you should be able to generate both 1 and 10).**
- 2. A static method called RandDouble() that accepts two integers and returns a random double between them. For this method you should be able to generate numbers such that $1 \leq x < 10$ for the method call RandDouble(1,10).**
- 3. Call your methods from another class without instantiating the class (i.e., call it just like Math.Random()) since your methods are defined to be static).**

using System;

class RandomHelper

{

public static int RandInt(int min, int max)

{

Random r = new Random();

return r.Next(min, max + 1);

}

public static double RandDouble(int min, int max)

{

Random r = new Random();

```
        return min + r.NextDouble() * (max - min);
    }
}

class Program
{
    static void Main()
    {
        Console.WriteLine("Random Integer : " + RandomHelper.RandInt(1, 10));
        Console.WriteLine("Random Double : " + RandomHelper.RandDouble(1,
10).ToString("F2"));
    }
}
```

OUTPUT:

Random Integer : 7

Random Double : 1.89